Week 10

Spring 2027 · Section 029 · MW 9:00–10:20 AM · RRC Room 8210

Course Level Objectives:

- Explain the role of probability in statistics. (CLO 4)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Relate the size of a p-value to the strength of evidence against H_0. (CLO 8) **2.** Compare a p-value to the significance level α to decide whether to reject H_0. (CLO 8) **3.** Write a two-step conclusion in context that avoids saying "accept H_0". (CLO 8)	e-Text: 4.3 Determining Statistical Significance Interactive Lecture: 4.3 Determining Statistical Significance Slides: 4.3 Determining Statistical Significance (includes "What is a p-value?" video)	Participation: P-values and Conclusions – resveratrol examples (p = 0.04, 0.12, 0.0003): strength of evidence, decision at α, conclusion in context Your Turn: Decide and Conclude	Online Homework: 4.3 Determining Statistical Significance (due Sun Apr 4, 11:59 PM) Take-Home Quiz 6 (due Mon Apr 5, 11:59 PM) – covers 4.1–4.3 (all objectives) Test 2 (Fri Apr 2 – Sun Apr 4, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Apr 1) (all objectives)	WileyPLUS StatKey Blackboard Respondus
4. Describe Type I and Type II errors in the context of a study. (CLO 8) **5.** Relate the significance level α to the probability of a Type I error. (CLO 4, CLO 8) **6.** Explain the trade-off between Type I and Type II errors and how sample size affects it. (CLO 8) **7.** Explain how multiple testing and publication bias produce false positives. (CLO 8) **8.** Distinguish statistical significance from practical significance. (CLO 8)	e-Text: 4.4 A Closer Look at Testing Interactive Lecture: 4.4 A Closer Look at Testing Slides: 4.4 A Closer Look at Testing (includes Khan Academy Type I/II video)	Participation: Errors in Testing – battery-life example (a)–(d): describe each error in context; courtroom analogy matching; vitamin E multiple-testing questions Your Turn: Errors and Multiple Tests (ambulance response times)	Online Homework: 4.4 A Closer Look at Testing (a) (due Sun Apr 4, 11:59 PM) Online Homework: 4.4 A Closer Look at Testing (b) (due Sun Apr 4, 11:59 PM) Take-Home Quiz 7 (due Mon Apr 12, 11:59 PM) – covers 4.4, 4.5, 5.1 (all objectives) Test 3 (Fri Apr 30 – Sun May 2, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Apr 29) (all objectives)	WileyPLUS StatKey Blackboard Respondus